

Chapter 15: Rectilinear Motion (Part 1)

Study Guide: Uniform Rectilinear Motion (URM)

Grade 10 Physics

1. Mathematical Prerequisite: The Straight Line

Before studying URM, it is essential to remember the mathematical equation of a straight line. The general form of a straight-line equation in math is:

$$y = a \cdot x + b$$

- **y-intercept (b):** The value of y when $x = 0$.
- **Slope (a):** Represents the steepness of the straight line.

2. Characteristics of URM

A motion is classified as **Uniform Rectilinear Motion (URM)** if it satisfies two main conditions:

Key Characteristics

- **Trajectory:** The path of the object is a **straight line**.
- **Velocity ($\vec{v}$):** The velocity vector is **constant** in magnitude, direction, and sense.
($v = \text{constant} = v_{\text{average}}$)
- **Acceleration ($\vec{a}$):** Because the velocity does not change, the acceleration is **zero**
($a = 0$).

3. Time Equations of Motion

In kinematics, we express the position x of an object as a function of time t . Because the $x(t)$ graph is a straight line, it mirrors the math equation $y = ax + b$.

General Form ($x_0 \neq 0$)

If the object does *not* start from the origin:

$$\begin{aligned}a &= 0 \\v &= \text{constant} \\x(t) &= v \cdot t + x_0\end{aligned}$$

Here, x_0 is the initial position at $t = 0$.

Starting from Origin ($x_0 = 0$)

If the object starts exactly at the origin O:

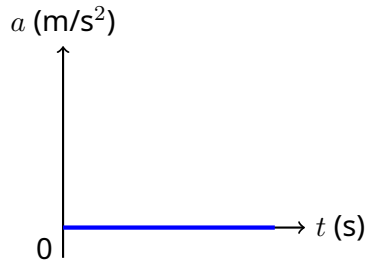
$$\begin{aligned}a &= 0 \\v &= \text{constant} \\x(t) &= v \cdot t\end{aligned}$$

Here, position x is directly proportional to time t .

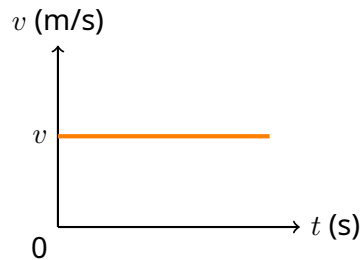
4. Graphs of Motion Quantities vs. Time

Understanding the graphical representation of URM is critical. Below are the three fundamental graphs for an object in Uniform Rectilinear Motion with a positive velocity and an initial position $x_0 > 0$.

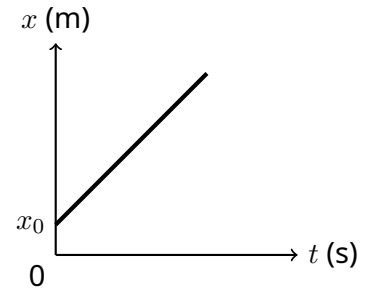
Acceleration-Time ($a - t$)



Velocity-Time ($v - t$)



Position-Time ($x - t$)



Graphical Note: The **slope** of the $x - t$ graph represents the **velocity** (v) of the object.

5. Application Problems

Problem 1: Guided Application

A car moves at a constant speed of 80 km/h on a rectilinear path.

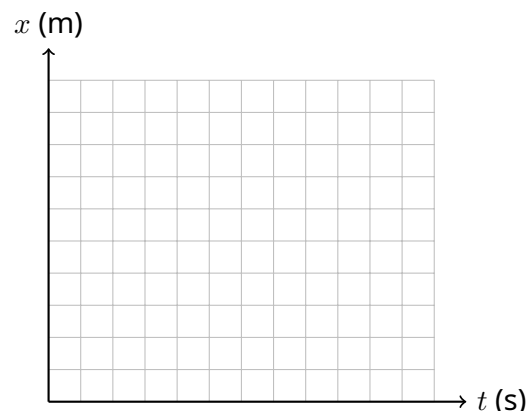
- What is the exact nature of this motion? Justify.
- Deduce the value of its acceleration a .
- Assuming the car starts its journey at the origin ($x_0 = 0$), write the equation representing its position x as a function of time t .
- How long does it take to travel 100 km?

Problem 2: From Data to Graph

The positions of a mobile at specific moments in time are recorded in the table below:

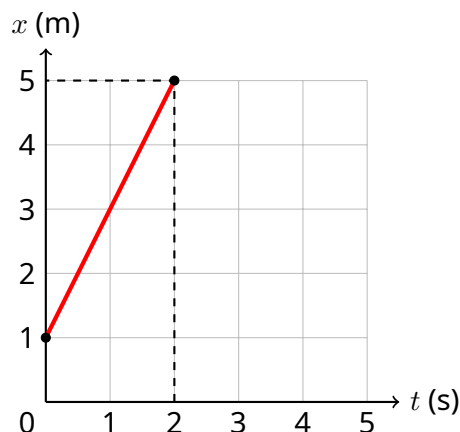
Time t (s)	0	1	2	3	4
Position x (m)	2	5	8	11	14

- Plot the graph of position x versus time t in the grid provided.
- Determine the equation of the plotted line graph.
- Deduce the original position x_0 and the speed of the mobile.
- What is the nature of the motion? Justify your answer.



Problem 3: From Graph to Characteristics

The graph below represents the variation of position x of a mobile with respect to time t .



- Determine graphically the initial position x_0 of the mobile.
- Calculate the slope of the graph. What physical quantity does it represent?
- Deduce the equation of motion $x(t)$.
- Conclude the nature of the motion.
- What is the position of the mobile at $t = 1$ s?

Problem 4: Multi-Phase Motion

A mobile moves along a straight path (x-axis) starting from the origin ($x_0 = 0$). The motion consists of three distinct phases:

- **Phase 1:** Moves in the positive direction with a uniform rectilinear motion (URM) at a speed of 4 m/s for 5 s.
- **Phase 2:** Stops at that position for a duration of 3 s.
- **Phase 3:** Reverses direction and returns to the origin ($x = 0$) with a constant speed in 4 s.

- Determine the position of the mobile at the end of Phase 1.
- State the velocity of the mobile during Phase 2.
- Calculate the velocity of the mobile during Phase 3. (Note the sign of the velocity).
- Plot the position-time graph ($x - t$) for the entire journey.

